

Universal Deployment & Submission Guide

Overview Building your API locally is only the first step of the software engineering lifecycle. To be evaluated, your code must survive in a production environment. For all project phases, you will deploy your API to **Render** (a free cloud hosting platform) and submit the live URL to the automated grading system.

Because Render uses a "free tier" architecture, your server will go to sleep after 15 minutes of inactivity. You must strictly follow the testing and submission timeline below to ensure the automated grader does not fail your submission due to a server timeout.

Step 1: Prepare Your Code for Production

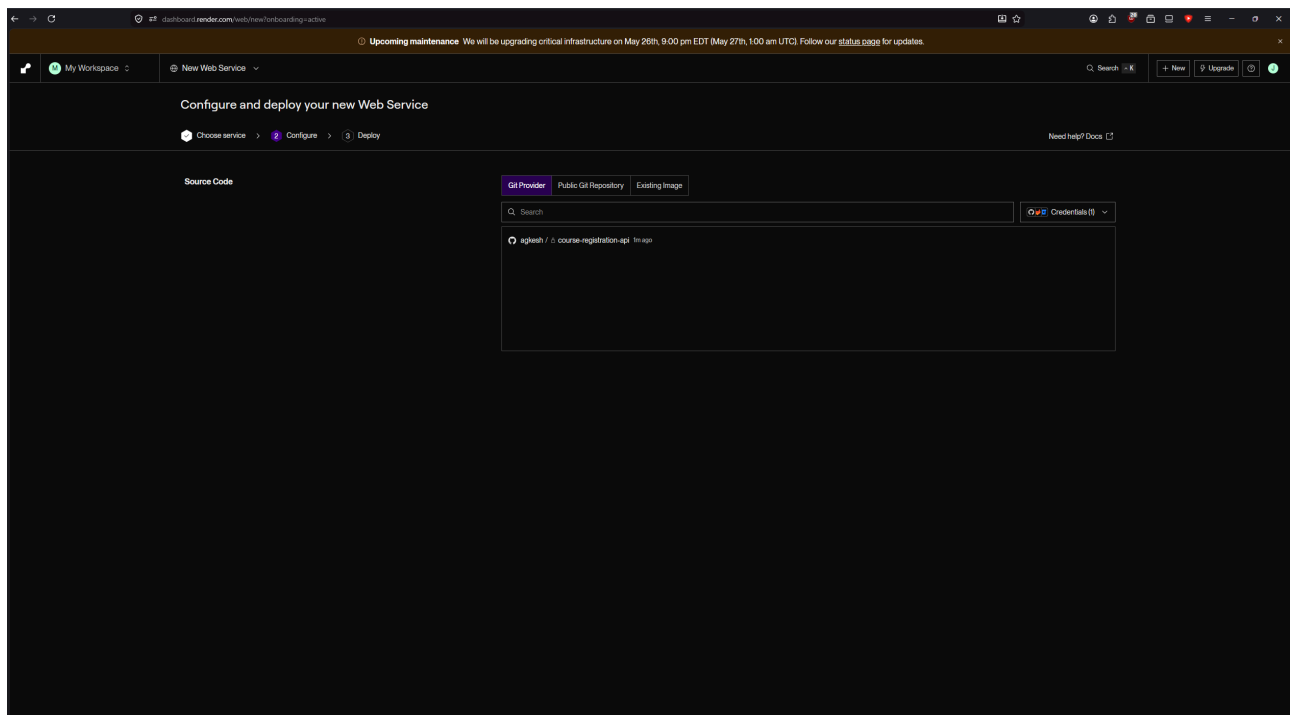
Cloud servers assign ports dynamically. If your code hardcodes a static port, it will crash on Render. Before pushing your final commit to GitHub, you must ensure your application listens to the environment's assigned port.

For Python (FastAPI): In your main file, ensure you are not hardcoding the port in `uvicorn.run()`. Alternatively, you can rely on Render's Start Command to pass the dynamically assigned environment port directly.

Step 2: Deploy to Render

You only need to configure Render once. After the initial setup, every time you push code to your GitHub `main` branch, Render will automatically deploy the newest version.

1. Go to [Render.com](https://render.com) and create a free account using your GitHub login.
2. Click **New +** and select **Web Service**.



3. Connect your GitHub account and select your `course-registration-api` repository.

4. Fill out the configuration settings exactly as follows:

- **Name:** `yourname-course-api` (or any unique name)
- **Region:** Ohio (US East) or closest available
- **Branch:** `main`
- **Instance Type:** Free

5. Set the Build and Start Commands:

- Build Command: `pip install -r requirements.txt`
- Start Command: `uvicorn main:app --host 0.0.0.0 --port $PORT`

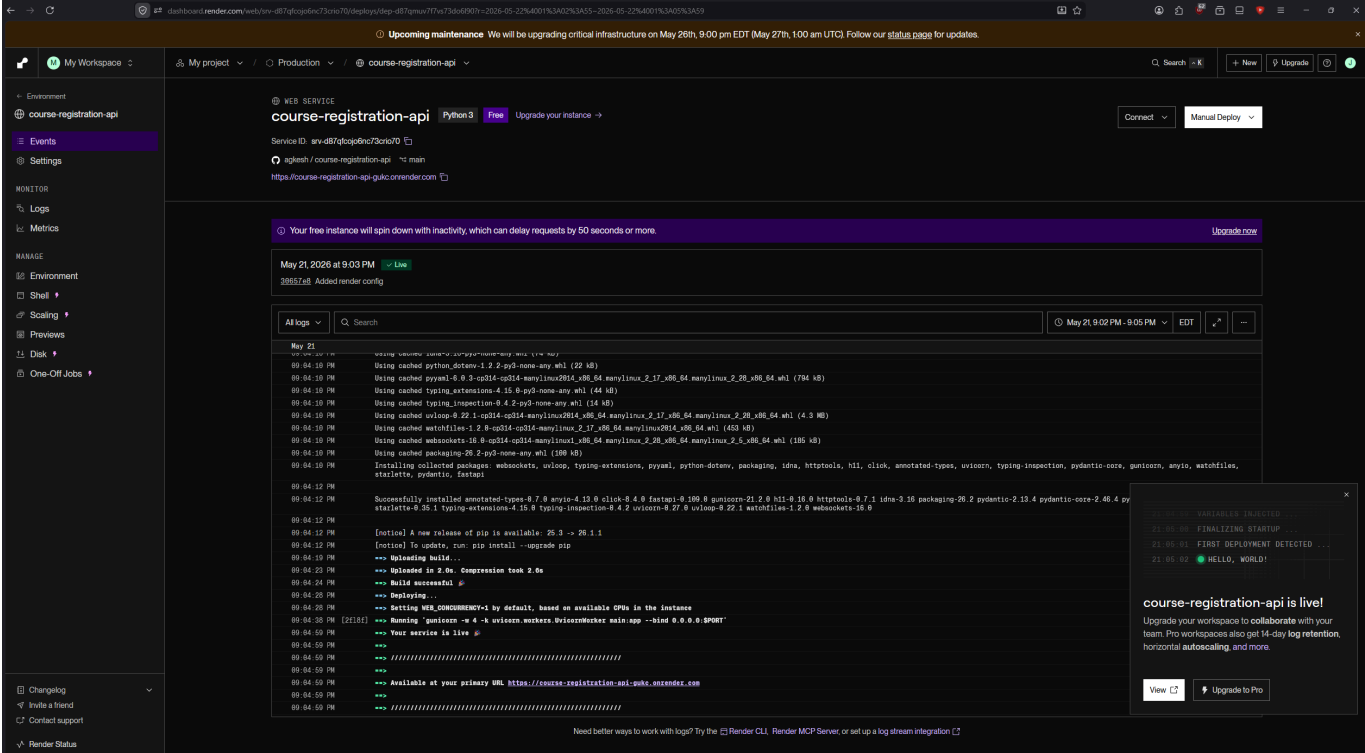
6. Click **Create Web Service**. Wait 2-3 minutes for the build logs to finish and display "Your service is live 🎉".

Step 3: Self-Testing and The "Cold Start" Wake-Up

Render provides you with a live Base URL (e.g., `https://yourname-course-api.onrender.com`).

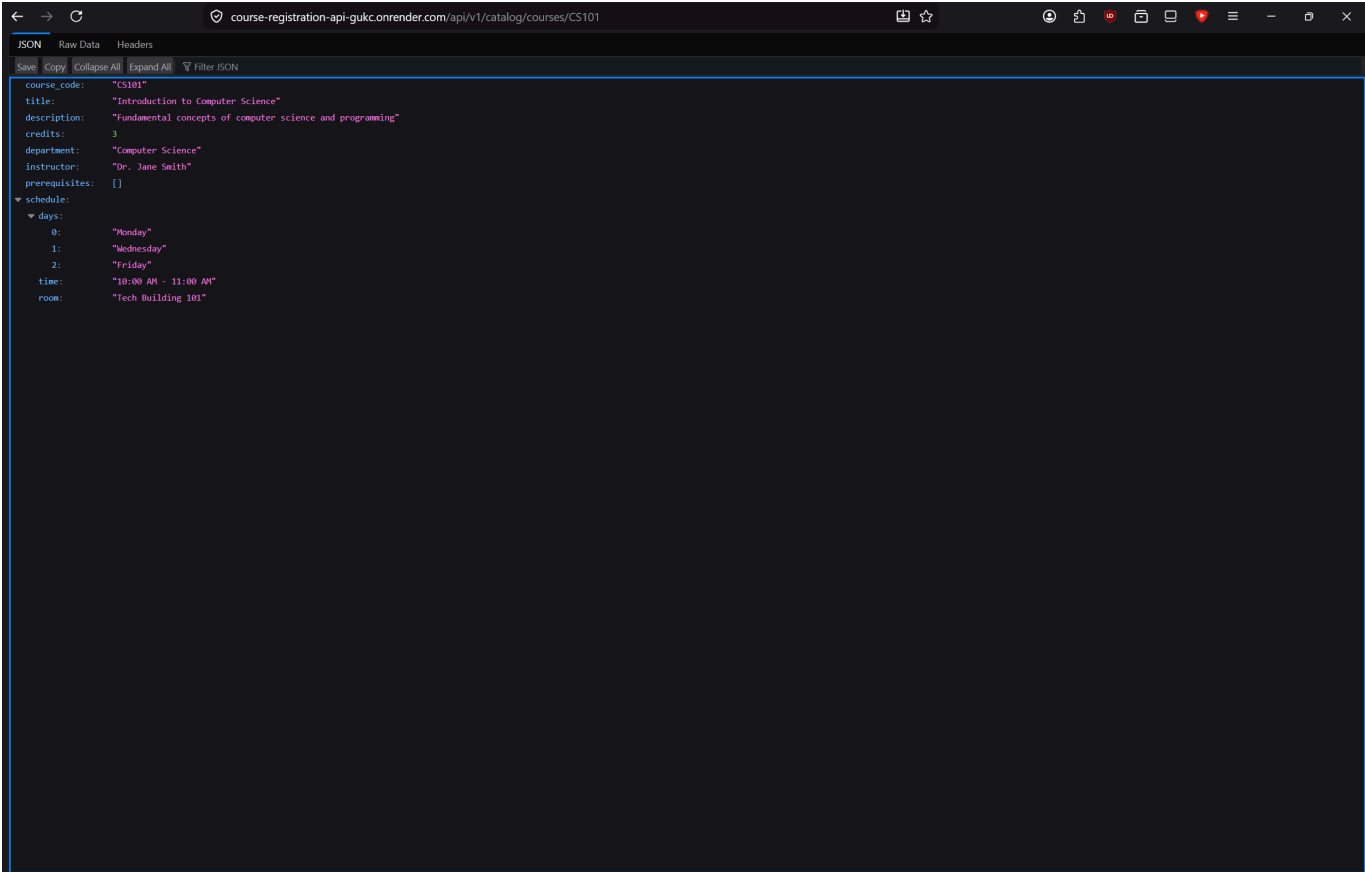
The Cold Start Problem: On Render's free tier, if your API receives no traffic for 15 minutes, it goes to "sleep" to save server resources. When the next request comes in, the server takes **30 to 60 seconds to wake up**.

If the automated grader hits a sleeping server, the grader will time out after 5 seconds and automatically assign you a failing grade. You must manually wake up your server before submitting.



How to self-test:

- 1. Open Postman or your web browser.
- 2. Send a standard GET request to one of your live endpoints (e.g., <https://yourname-course-api.onrender.com/api/v1/catalog/courses/COSC3506>).
- 3. If the server was asleep, this request will hang for up to a minute. Wait for it.
- 4. Once you receive a 200 OK response, your server is officially **awake** and running at full speed.
- 5. Send one more request to verify it responds instantly.



Step 4: The 15-Minute Submission Window

Once your server is awake, a 15-minute countdown begins. If it receives no traffic for 15 minutes, it will go back to sleep.

1. **Immediately** after verifying your API is awake and responding instantly, copy your Base URL (e.g., <https://yourname-course-api.onrender.com>). Do not include trailing slashes or specific endpoint paths.
2. Log into the course grading system.
3. Paste your Base URL into the submission box and click **Submit**.
4. The auto-grader will execute its test suite against your API immediately upon submission.

Critical Rule: If you test your API, go make a coffee, come back 20 minutes later, and click Submit, the grader will hit a sleeping server and fail you. **Always ping your API immediately before clicking Submit.**